

Homework 05

CSCE 310

Instructions

This assignment consists of 5 analytical problems and 2 programming problems. Your solutions to the analytical problems must be submitted, as one PDF (with your last name and first name). While handwritten (then scanned) solutions to the analytical problems are acceptable, you are strongly encouraged to typeset your solutions in L^AT_EX or a word processor with an equation editor. The legibility of your solutions is of great importance.

Programming Assignment

You have the option to use either Python or C++ for the programming assignment. Your methods will be tested on the cse-linux-01.unl.edu server, so make sure that they are runnable on that server.

For C++, we will be using g++ version 7.5.0 and be compiled for C++ 2014.

Example: `g++ *.cpp -std=c++14` then run the executable file

For python, just run the command: `python *.py`

You will submit a zipped file named **Homework5_lastName_firstName** that consists of: OurCSCE310Tree.h and OurCSCE310Tree.cpp for C++ or OurCSCE310Tree.py, along with your PDF, via **Canvas**. You can add a readme file including the command that you use to compile your code.

rotateLeft

`rotateLeft` is a function that will rotate a `OurCSCE310Tree` to the left.

rotateRight

`rotateRight` is a function that will rotate a `OurCSCE310Tree` to the right.

deleteNode (15 **Points Extra Credit**)

`deleteNode` is a function that will receive a number and (if it exists in the `OurCSCE310Tree`) delete it. You do not need to rebalance the tree if it becomes unbalanced after a deletion.

Question 1 (10 points)

Question 11.2.10, Parts *b* through *e*, in *The Design and Analysis of Algorithms*

Question 2 (10 points)

Consider the problem of searching for the pattern “band” in the text “bananaband” using Horspool’s Algorithm.

1. Construct a shift table for the pattern “band”.
2. Apply Horspool’s Algorithm to allocate the pattern “band” in the text “bananaband”.
3. How many comparisons Horspool’s Algorithm needed versus the naive brute force algorithm.

Question 3 (10 points)

Question 11.3.2 in *The Design and Analysis of Algorithms*

Question 4 (10 points)

Question 11.3.6 in *The Design and Analysis of Algorithms*

Question 5 (10 points)

Question 11.3.11 in *The Design and Analysis of Algorithms*

Point Allocation

Question	Points
Question 1	10
Question 2	10
Question 3	10
Question 4	10
Question 5	10
rotateLeft	
Test Cases	1 × 15
Compilation	10
rotateLeft Total	25
rotateRight	
Test Cases	1 × 15
Compilation	10
rotateRight Total	25
Total	100